BT301 SUBJECTIVES

FINAL TERM- 100%

Introduction-to-Biotechnology

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SUBJECTIVES

Difference between pBR322 and pBR325:

pBR322 and pBR325 are plasmids used in genetic engineering. pBR322 is an in vitro constructed cloning vehicle with ApR and TcR genes from RSF2124 and pSC101, and a replication element from pMB1. It has 40 restriction enzyme target sites but is less suitable for long DNA segments. In contrast, pBR325, a derivative of pBR322, also encodes chloramphenicol resistance along with ampicillin and tetracycline, offering expanded applications and improved vectors like pUC19 and pAT153. These plasmids have been crucial in DNA cloning and genetic engineering experiments, contributing to advancements in molecular biology research.

What is the cosmid vector:

A cosmid vector is a type of DNA carrier that includes a specific segment from a bacteriophage called lambda (ʎ). Including “Cos” site. These vectors, called cosmids, are useful for cloning large sections of foreign DNA. They work well with an in vitro packaging system to insert DNA fragments. Cosmids are especially useful for creating libraries of genetic material from eukaryotic organisms.

Write a note on RT-PCR:

RT-PCR, or Reverse Transcription Polymerase Chain Reaction, is a technique used to amplify and detect RNA molecules. It involves converting RNA into DNA using reverse transcriptase and then amplifying the DNA using PCR. RT-PCR is widely used in research, diagnostics, and forensic analysis for studying gene expression, detecting viruses, and genetic testing.

Gauecher disease:

Gaucher disease is a rare genetic disorder where glucocerebroside accumulates in cells due to a deficiency of the enzyme glucocerebrosidase. This buildup leads to organ and tissue damage. Symptoms include enlarged liver and spleen, anemia, and bone abnormalities. Enzyme replacement therapy is a common treatment.

Chemical composition of DNA:/ Note on DNA

DNA, or deoxyribonucleic acid, serves as the genetic blueprint for living organisms. Its primary function is to store, transmit, and express genetic information. DNA encodes the instructions for building and maintaining an organism's structure, controlling its growth, development, and functioning. DNA, or deoxyribonucleic acid, is composed of nucleotide units. Each nucleotide consists of a phosphate group, a deoxyribose sugar molecule, and one of four nitrogenous bases: adenine (A), thymine (T), cytosine (C), or guanine (G). The sequence of these bases forms the genetic code. Adenine pairs with thymine, and cytosine pairs with guanine through hydrogen bonds, creating the double-stranded helical structure of DNA. This unique composition allows DNA to store and transmit genetic information, playing a fundamental role in heredity and cellular processes.

Whats happen during Chromosome jumping:

During chromosome jumping, the targeted DNA is first identified, cut into fragments using restriction enzymes, and then circularized. This process facilitates the proximity of DNA sequences that were originally distant in the genome. The resulting circularized DNA fragments, originating from the closure sites, constitute a jumping library. Chromosome jumping is applied in molecular biology techniques like the cloning of genes, exemplified by the cloning of the human cystic fibrosis gene, utilizing the ability to bring together disparate DNA sequences for comprehensive genomic analysis.

How can PCR help in improving quality of environment?

1. PCR detects environmental pollutants like heavy metals and pathogens.
2. It monitors microbial diversity in various habitats.
3. PCR studies genes involved in bioremediation for effective pollutant removal.
4. Environmental DNA (eDNA) analysis identifies species presence and biodiversity.
5. PCR tracks changes in gene expression due to environmental stressors.
6. Overall, PCR aids in environmental monitoring, assessment, and management, ensuring ecosystem preservation and human health protection

Genome mapping and its type?

Assigning of a specific gene to particular region of a chromosome and determining the location of and relative distances between genes on the chromosome. There are two type here:1= Genetic Linkage Mapping: Illustrates the order and relative distances between genes or genetic markers on chromosomes. 2= Physical Mapping: Provides actual distances between landmarks on a chromosome, typically measured in base pairs.

Significant of PCR in medical field / Role of PCR in Medical:

PCR has major impact on medicine especially in the field of clinical microbiology or diagnosis. Molecular tools have also allowed to perform prenatal genetic diagnosis. PCR enables the detection of genetic disorders such as Down syndrome, cystic fibrosis, and sickle cell anemia during pregnancy.

What is biopaninng?

The primary screening method involves enriching phage clones with binding affinity for the target through a process called biopanning. Biopanning is a process used to find specific molecules, like antibodies, by enriching them from a mixture.

DNA sequence databases:

Large amounts of sequence data accumulated since the development of current DNA sequencing technology are maintained in three databases: i) National Center for Biotechnology Information ii) DNA Databank of Japan iii) European Bioinformatics Institute-UK

Eukaryotic DNA types:

1= Protein Coding Genes, 2= Tandemly Repeated Genes, 3= Repeated DNA, 4= Unclassified Spacer DNA.

Protein Coding Genes can further be classified as: 1= Duplicated and Diverged Genes, 2= Functional Gene Families and Non-functional.

Pseudogenes Tandemly Repeated Genes encode: 1= rRNA, 2= 5sRNA, 3= tRNA, 4= Histones

Multiplex and Real time PCR:

Multiplex PCR: Multiplex PCR is a variant of PCR which enables simultaneous amplification of many targets of interest in one reaction by using more than one pair of primers. The use of multiple, unique primer sets within a single PCR reaction to produce amplicons of varying sizes specific to different DNA sequences.

Real-time quantitative PCR (qPCR): qPCR is used to measure the quantity of a target sequence. It is used to measure starting amounts of DNA, cDNA, or RNA. In real-time PCR, the accumulation of amplification product is measured as the reaction progresses, in real time, with product quantification after each cycle. Q-PCR is commonly used to determine whether a DNA sequence is present in a sample and the number of its copies in the sample.

Disadvantages of RT PCR:

1= It can be costly because of needing special equipment and materials. 2= It's a bit complex to set up and analyze the data, requiring know-how. 3= It's sensitive to stuff in the samples that can mess up the results. 4= It might struggle with accurately measuring very low or high levels of stuff in the sample. 5= Making sure the data is compared fairly across different samples can be tricky. 6= It's not great at testing many things at once. 7= Sometimes, it might give wrong results if there's contamination.

Basic concept of Maxam Gilbert DNA sequencing Method:

The Maxam-Gilbert method is a DNA sequencing technique that uses chemical reagents to break the DNA at specific bases. By inducing base-specific cleavage, this method allows researchers to determine the sequence of nucleotides in a DNA molecule.

Write a note on shotgun sequencing:

Shotgun sequencing is a method used to sequence long strands of DNA. 1= Breaking Down DNA: In shotgun sequencing, long DNA strands are broken into smaller fragments. 2= Individual Sequencing: Each fragment is sequenced individually, allowing for efficient processing. 3= Reassembly Process: After sequencing, the fragments are reassembled in their original order. 4= Efficient Genome Sequencing: This method is effective for sequencing large genomes, aiding in the study of genetics and genetic disorders.

Application of PCR:

PCR has widespread applications in various fields of life sciences including genetic engineering, medical, forensic, agriculture, environment etc.

Benefits of cDNA in recombinant DNA technology:

Here are the benefits of using cDNA in recombinant DNA technology: 1= Simplified Cloning: cDNA lacks non-coding regions, making it easier to clone genes into vectors for various applications. 2= Focused Gene Expression: It represents actively expressed genes, allowing researchers to study specific gene functions more effectively. 3= Efficient Protein Production: cDNA simplifies the process of producing proteins in host cells, as it contains only coding sequences. 4= Amplification of Low-Abundance Genes: cDNA synthesis can amplify rare or low-abundance transcripts, aiding in the study of less common genes. 5= Discovery of New Genes: cDNA libraries help in the discovery of new genes and understanding their roles in biological processes.

How comparative genomics find conserved gene in different species:

Comparative genomics identifies conserved genes across species by analyzing DNA sequences from different organisms. Computational algorithms align genomic data, revealing regions with high sequence similarity, suggesting functional importance. Conserved genes are evolutionarily retained due to their crucial roles, indicating shared biological functions. Comparative analysis identifies conserved motifs, regulatory elements, and coding sequences, aiding in understanding gene function, evolution, and potential therapeutic targets. The conservation of genes provides insights into fundamental biological processes and highlights key elements shaping the diversity of life.

Define Chain termination or deoxy procedure:

The chain termination or deoxy method, also known as the Sanger method, was developed by Sanger et al. (1977). In this technique, DNA polymerase incorporates dideoxynucleoside triphosphates (ddNTPs) as substrates. These ddNTPs act as chain terminators, stopping DNA synthesis. The procedure involves carrying out DNA synthesis in the presence of all four deoxynucleoside triphosphates (dNTPs), one of which may be labeled with 32P, in separate incubation mixes. As the ddNTPs randomly terminate DNA synthesis at different positions, a series of fragments with varying lengths are generated, allowing determination of the DNA sequence.

What are the Feature of D loop of human mitochondrial genome:

The D-loop, or displacement loop, in the human mitochondrial genome is a non-coding region where replication and transcription of mitochondrial DNA begin. It contains regulatory elements essential for controlling gene expression and replication. Additionally, the D-loop is highly variable among individuals and is often used in population genetics and forensic studies for tracing maternal ancestry and identifying individuals.

Explain harshy marthachase experiment of discovery of DNA

The Hershey-Chase experiment (1952) confirmed that DNA, not protein, carries genetic information. Martha Chase and Alfred Hershey used T2 bacteriophage labeled with radioactive sulfur (protein) or phosphorus (DNA). Only the phosphorus-labeled DNA entered Escherichia coli bacteria, leading to viral replication. This demonstrated that DNA, not protein, serves as the hereditary material, solidifying the understanding of DNA's role in genetics and molecular biology.

Benton and Davis lift procedure:

Benton and Davis (1977) devised a method called plaque lift, in which nitrocellulose filter is applied to the upper surface of agar plates, making direct contact between plaques and filter. Number of alternative labelling methods are available that avoid the use of radioactivity. It included digoxigenin or biotin.

Automated DNA sequencing:

To automate the process, it is desirable to acquire sequence data in real time by detecting the DNA bands within the gel during electrophoretic separation.

SNPs as Physical Markers:

Single nucleotide polymorphisms (SNPs) are variations occurring at single base-pair positions in genomic DNA within a population. SNPs are highly significant in physical mapping of genomes due to their prevalence and importance as genetic markers. Various methods have been developed to detect SNPs, including enzymatic, electrophoretic, solid-phase, and chromatographic analysis.

Applications of PCR:

Testing of genetic disease mutations, Monitoring the gene in gene therapy, Detecting disease-causing genes in the parents.

PCR is a common tool used in medical and biological research labs. It is used in the early stages of processing DNA for sequencing?, for detecting the presence or absence of a gene to help identify pathogens? during infection, and when generating forensic DNA profiles from tiny samples of DNA.

Role of genome sequencing:

Genome sequencing provides insights into genetic variation, disease causes, and drug responses. It aids in understanding evolutionary relationships, functional genomics, and personalized medicine. It enables precise diagnostics, facilitates crop improvement, and contributes to advancements in biotechnology and medical research.

Four type of cloning strategy:

1= Gene Cloning: Isolating and replicating a specific gene.
2= Reproductive Cloning: Generating an organism with identical genetic material.
3= Therapeutic Cloning: Producing embryonic cells for medical purposes.
4= Molecular Cloning: Creating multiple copies of DNA fragments for analysis or manipulation in molecular biology research.

AFLP PCR:

AFLP-PCR, or Amplified Fragment Length Polymorphism Polymerase Chain Reaction, is a technique used to detect differences in DNA fragments caused by single nucleotide polymorphisms (SNPs) or insertions/deletions (INDELs). AFLP is a PCR-based tool commonly utilized in genetic engineering and molecular biology research to analyze genetic variation and identify molecular markers associated with traits of interest.

Define Histones & its types:

Histones are proteins found in eukaryotic cell nuclei that package and organize DNA into structural units called nucleosomes.

There are five main types of histones: 1= Histone H1: Binds to the linker DNA between nucleosomes and helps stabilize the chromatin structure. 2= Histone H2A: Forms part of the nucleosome core structure and plays a role in regulating gene expression. 3= Histone H2B: Also part of the nucleosome core structure and involved in gene regulation. 4= Histone H3: Forms the central core of the nucleosome and is important for chromatin compaction, 5= Histone H4: Along with H3, makes up the central core of the nucleosome and plays a role in chromatin structure and stability.

RAPD (Random Amplification Polymorphic DNA):

Random amplification of polymorphic DNA is a type of PCR reaction but the segments of DNA that are amplified are random.

Principle: RAPD utilizes short, random oligonucleotide primers to anneal to multiple sites within the genomic DNA. During PCR, these primers amplify random DNA segments, resulting in a complex mixture of fragments.

Procedure: 1= DNA Extraction: Genomic DNA is isolated from samples of interest. 2= PCR Amplification: The RAPD reaction mixture contains template DNA, random primers, DNA polymerase, nucleotides, and buffer. PCR amplification involves denaturation, primer annealing, and extension steps. 3= Gel Electrophoresis: Amplified DNA fragments are separated by size using agarose gel electrophoresis, followed by staining to visualize the bands. 4= Band Interpretation: The presence or absence of bands in the gel indicates genetic variation among samples.

What are the Applications of RAPD:

1= RAPD finds applications in: Genetic fingerprinting for individual identification. 2=Population genetics to study genetic diversity and structure. 3= Phylogenetic analysis to infer evolutionary relationships. 4= Genetic mapping for marker-assisted breeding. 5= Disease diagnosis by identifying associated markers.

Advantages: RAPD is rapid, cost-effective, highly polymorphic, versatile, and requires minimal DNA input.

What is interrupted matting?

Interrupted mating is a genetic technique used to map genes on a bacterial chromosome by stopping the transfer of genetic material during conjugation at different time intervals. It helps determine the sequence and relative distances between genes on the chromosome, aiding in the construction of genetic maps and the study of bacterial genetics.

Importance of stability of plasmid and Effects of copy number of plasmid.

Importance of Stability of Plasmid • Vital for maintaining genes of interest • Ensures consistent gene expression • Enables long-term storage • Crucial for reliable experimental results • Necessary for continuous bioproduction processes • Important for biosafety considerations • Prevents loss of genetic material during bacterial growth and replication • Ensures integrity and functionality of genetic constructs.

The effect of plasmid copy number:

1= The numbers of ribosomes in a cell far exceed any one class of m RNA, 2= One way of increasing the expression of cloned gene is to increase the numbers of corresponding transcript.

Principles of western blotting:

Western blotting involves separating proteins by electrophoresis, transferring them to a membrane, and probing with specific antibodies to detect target proteins. This technique enables the identification and quantification of proteins based on their size and antibody binding.

What is primer Extension? Explain single method:

Primer extension is a technique used in molecular biology to copy a specific segment of DNA In the single primer method of primer extension, a short piece of chemically made DNA (7-20 base pairs long) is used to start copying a longer piece of DNA. This short piece has one base that's different from the original DNA. After putting this copied DNA into E. coli bacteria, we get two types of DNA: one that's exactly like the original and one with the changed base. This copied DNA can be used for various molecular biology applications such as cloning or genetic analysis.

Why PCR is considered as alternative of DNA genome:

PCR is considered an alternative to traditional methods for studying DNA genomes because it's fast, sensitive, specific, versatile, and can be automated. It rapidly amplifies target DNA sequences, detects small amounts of DNA, and selectively amplifies specific regions without the need for cloning. PCR's efficiency and automation make it invaluable for various applications in genetics, molecular biology, and diagnostics.

Define Chromosome walking?

Chromosome walking is an approach used to find gene sequences whose function is unknown but their genetic location is known. It starts with a cloned genomic fragment as a starting point. For example, in the human genome, this starting point could be a sequence closely linked to a disease locus. One drawback of this method is that each DNA segment used must not be repeated elsewhere in the genome.

Define FISH?

Fluorescence in situ hybridization (FISH) is a cytogenetic technique that employs fluorescent probes. These probes specifically bind to complementary sequences on chromosomes, allowing for the visualization and mapping of specific DNA sequences within the cell.

Define Spotted DNA arrays:

Spotted DNA arrays are created by printing DNA samples onto specially treated microscope slides. This process involves transferring or spotting DNA clones or PCR products onto a solid support, where they become immobilized. These arrays are used in functional genomics and proteomics studies for various applications, including analyzing DNA binding interactions and studying gene expression patterns.

STS in physical maps?

STS stands for Sequence Tagged Site. In physical mapping, STSs are short DNA sequences with unique locations in the genome. They serve as landmarks to which other DNA sequences can be anchored. By identifying the presence or absence of specific STSs in a set of clones, researchers can order and orient those clones along the chromosome. STSs are commonly used in constructing physical maps of genomes, helping to determine the order, distance, and organization of DNA sequences.

What is biotechnology and waited names of it's various branches?

Biotechnology is the use of microbes, animal/plant cells and their products to synthesize, break down or transform materials. Primarily it includes the use of recombinant DNA technology and genetic engineering techniques to improve upon the quality of processes.

1. Medical Biotechnology
2. Agricultural Biotechnology
3. Industrial Biotechnology
4. Environmental Biotechnology
5. Forensic Biotechnology

Define primary, secondary, tertiary structure of protein:

Primary structure involves the linear sequence of amino acids in a polypeptide chain. Secondary structure results from interactions such as alpha-helices and beta-sheets, folding due to hydrogen bonding. Tertiary structure represents the overall three-dimensional arrangement of a single polypeptide chain. Quaternary structure involves the interaction of multiple polypeptide chains in a protein complex. These hierarchical levels collectively determine a protein's function and are crucial for its stability and biological activity.

DNA is polymer of deoxyribonucleotide. Enlist and explain the chemical composition of deoxyribonucleotide:

A deoxyribonucleotide, the monomer of DNA, consists of three components: a deoxyribose sugar molecule, a phosphate group, and one of four nitrogenous bases (adenine, thymine, cytosine, or guanine). The deoxyribose sugar forms the backbone, linked by phosphodiester bonds between adjacent nucleotides. The nitrogenous base, either a purine (adenine or guanine) or pyrimidine (thymine or cytosine), connects to the sugar. This composition imparts the unique sequence and structure to DNA, encoding genetic information and facilitating various cellular processes.

Yeast libraries and prokaryotes libraries:

Aspect	Yeast Libraries	Prokaryotic Libraries
Organism	Yeast (Saccharomyces cerevisiae)	Bacteria (e.g., Escherichia coli)
Cell Type	Eukaryotic, with nucleus and complex cellular machinery	Prokaryotic, lacking nucleus and simpler cellular structure
Applications	Cloning, functional genomics, gene expression studies	Cloning, microbial genetics, protein production
Complexity	More complex due to eukaryotic cellular processes	Simpler due to absence of nucleus and organelles
Suitable for	Studying eukaryotic gene regulation, protein interactions	Studying microbial genomes, gene expression, and biotech

What is principal strategy used in siRNA technology ?

SiRNA Technology: Small interfering RNA (siRNA) is a widely used RNA interference (RNAi) tool. It induces short-term silencing of protein-coding genes by targeting specific mRNA molecules for degradation. SiRNA is a synthetic RNA duplex designed to match the sequence of the target mRNA. It has significant applications in gene silencing and holds promise as a therapeutic tool in various fields, including medicine and biotechnology.

Role of electroporation technique:

Electroporation is a rapid and straightforward method for introducing cloned genes into diverse microbial, plant, and animal cells, such as E. coli. This technique involves applying an electric field to cells, temporarily increasing the permeability of the cell membrane. This heightened permeability allows the uptake of DNA, facilitating efficient gene transfer and making electroporation a valuable tool in genetic engineering and molecular biology applications across various organisms.

What is difference between circular DNA and Open circle or OC DNA?

Circular DNA is a complete, covalently closed loop, while open circle (OC) DNA is a partially cleaved circular DNA with a break or nick in one strand. OC DNA results from enzymatic or chemical treatments and is distinct from the continuous, closed structure of circular DNA.

Chromosome walking?

Chromosome walking is an approach used to find gene sequences whose function is unknown but their genetic location is known. It starts with a cloned genomic fragment as a starting point. For example, in the human genome, this starting point could be a sequence closely linked to a disease locus. One drawback of this method is that each DNA segment used must not be repeated elsewhere in the genome.

What is use of RH mapping?

Radiation hybrid (RH) mapping RH mapping, or radiation hybrid mapping, is a technique used in genetics to map the locations of genes on chromosomes. It involves subjecting cells to radiation to induce breaks in the chromosomes, followed by fusing these damaged cells with cells from another species that are resistant to radiation. By analyzing the patterns of retention and loss of genetic markers or sequences from the original chromosomes, researchers can determine the order and distances between these markers. RH mapping is valuable for constructing high-resolution maps of genomes, identifying disease-associated genes, studying genome evolution, and facilitating comparative genomics across species.

Components of Bacterial genome:

Bacterial genomes are characterized by high coding density, with 85-90% of DNA encoding genes. Prokaryotes exhibit a gene frequency of 1 gene per 1000 bases. A bacterial genome consists of a single circular DNA molecule containing essential components. The chromosome carries genes for vital cellular functions, and plasmids, additional circular DNA, may harbor accessory genes. Regulatory elements control gene expression, ensuring proper cellular processes. Bacterial genomes lack introns, and operons organize related genes. Mobile genetic elements like transposons contribute to genetic diversity. Understanding these components is crucial for studying bacterial genetics, evolution, and developing therapeutic interventions.

Describe significance of inverse PCR?

Inverse PCR is a valuable technique used to amplify unknown DNA sequences adjacent to known regions. Its significance lies in its ability to amplify DNA fragments with known sequences at both ends, allowing for the identification of flanking sequences or regions of interest. This method is particularly useful for mapping insertions, deletions, or rearrangements in genomes, studying gene regulation, identifying transposon insertion sites, and facilitating genome walking. Overall, inverse PCR helps elucidate genomic structure and function, aiding in various genetic and molecular biology studies.

Typelll restriction modification system:

Type III: Type III enzymes have symmetrical recognition sequences but otherwise resemble type I systems and are of little value.

Define Palindromic sequence:

A DNA locus whose 5'-to-3' sequence is identical on each DNA strand. The sequence is the same when one strand is read left to right and the other strand is read right to left. Recognition sites of many restriction enzymes are palindromic.

Define PCR-DNA profiling:

DNA profiling or DNA fingerprinting is a forensic technique used to identify individuals by characteristics of their DNA.

Chargaff’s rule:

Chargaff's rule, formulated by biochemist Erwin Chargaff, states that in DNA, the amounts of adenine (A) are equal to thymine (T), and the amounts of guanine (G) are equal to cytosine (C). This base pairing specificity is fundamental to the double-stranded nature of DNA, as adenine pairs with thymine via two hydrogen bonds, and guanine pairs with cytosine via three hydrogen bonds, ensuring structural stability and accurate replication of genetic information.

Alpha beta sheet Common in secondary structure protein?

Alpha helix and beta sheet are common secondary structures in proteins. The alpha helix is a coiled structure formed by twisting the polypeptide chain, while the beta sheet is a sheet-like structure formed by extended segments of the chain. These structures are stabilized by hydrogen bonds and contribute to the overall shape and stability of proteins.

Chloroplast gene function

Chloroplast genes encode proteins, tRNAs, and rRNAs essential for chloroplast function in photosynthesis. They contribute to the synthesis of pigments, electron transport chain components, and enzymes, playing a crucial role in energy conversion and the production of organic molecules in plant cells.

Comparative genomics of eukaryotes?

The minimal eukaryotic genome is smaller than many bacterial genomes, Comparative genomics can be used to identify genes and regulatory elements, It gives insight into the evolution of key proteins

What are the Screening strategies?

Major screening strategies involve 1). Genetic methods 2). Sequence-dependent screening 3). Screening expression libraries.

Genetic variation:

Genetic variation refers to the diversity of genetic material within a population or species. It arises from differences in DNA sequences, including mutations, recombination, and other genetic processes, influencing individual traits and contributing to the evolution and adaptation of organisms.

Requirement for expression in E. coli:

To express a functional protein in E. coli, the gene of interest is cloned into an expression vector. Efficient transcription and translation of the gene are necessary, along with codon optimization for E. coli. Some proteins may require post-translational modifications, and compartmentalization within the cell may be needed for proper functionality.

What is DNA and its type?

DNA, or deoxyribonucleic acid, is a molecule found in all living organisms that carries genetic information. It consists of two long chains of nucleotides twisted into a double helix structure. Each nucleotide contains a sugar (deoxyribose), a phosphate group, and one of four nitrogenous bases: adenine (A), thymine (T), cytosine (C), or guanine (G). The sequence of these bases along the DNA molecule determines the genetic instructions for an organism's development, functioning, and traits.

Types: 1= Genomic DNA 2= Mitochondrial DNA 3= Chloroplast DNA.

DNA Microarrays DNA Microarrays are utilized to simultaneously study the expression of numerous genes. The process involves immobilizing thousands of gene sequences on a glass slide, which are then detected through complementary base pairing between the sample and the gene sequences on the chip.

Profiling in human disease:

Profiling in human disease involves using arrays to study transcriptional profiles associated with various diseases. This approach helps identify novel disease markers and potential new drug targets.

Why the product of PCR varying in quantity:

The quantity of PCR product can vary due to factors like the amount of initial DNA template, PCR efficiency influenced by primer design and cycling conditions, potential PCR inhibition from contaminants, and variations in experimental technique and conditions.

What may be advantages of cloning in high capacity vector?

1= Larger Insert Size 2= Improved Gene Expression 3= Enhanced Genetic Engineering.

Work of Franklin and Winlikns?

Rosalind Franklin and Maurice Wilkins contributed to the discovery of the DNA double helix. Franklin's X-ray diffraction images provided critical insights into DNA's structure, and Wilkins collaborated with James Watson and Francis Crick, leading to the elucidation of the DNA model. Their work significantly advanced molecular biology and our understanding of genetics.

How many amino acids are present and give names of 5 amino acids?

There are 20 amino acids that serve as the building blocks of proteins. Five of them are:

1. Alanine (Ala): A nonpolar amino acid.
2. Valine (Val): A branched-chain amino acid.
3. Leucine (Leu): Another essential branched-chain amino acid.
4. Glutamine (Gln): A polar, uncharged amino acid.
5. Lysine (Lys): A positively charged amino acid.

Types of RNA:

RNA, or ribonucleic acid, comes in various types with distinct roles. Messenger RNA (mRNA) carries genetic information from DNA to ribosomes for protein synthesis. Transfer RNA (tRNA) transports amino acids to ribosomes during protein assembly. Ribosomal RNA (rRNA) is a structural component of ribosomes. Small nuclear RNA (snRNA) aids in RNA splicing, and small nucleolar RNA (snoRNA) guides chemical modifications in ribosomal RNA.

Protein Trafficking In Bacteria:

Proteins synthesized in the cytoplasm of bacteria often need to be transported to other cellular locations, such as the inner membrane, periplasmic space, or outer membrane. This process involves several mechanisms: General Secretion Pathway: Moves proteins from the cytoplasm across the cytoplasmic membrane into the periplasmic space for secretion outside the cell. ABC Pathway for Protein Export: Relies on ATPbinding cassette (ABC) transporters to move proteins across the cytoplasmic membrane using energy from ATP hydrolysis. Type II Secretion Pathway: Translocate proteins from the periplasmic space across the outer membrane. Proteins are first translocated into the periplasmic space by the Sec system before being exported through the outer membrane.

How functional complementation restores the lost function in higher organism like animals?

Functional complementation is the process by which a particular DNA sequence compensates for a missing function in a mutant cell and thus restores the wild type phenotype. Functional complementation is also possible in transgenic animals and plants. It has been used for complementation in transgenic mice to isolate the Shaker-2 gene.

What is one-way method for genome sequencing?

One-way genome sequencing involves reading DNA in a single direction, typically by sequencing only one strand of the double-stranded DNA. This method provides information on the sequence of one of the DNA strands, offering insights into genetic information.

What happens when super coil DNA molecules receive nick:

When super-coiled DNA molecules receive a nick, some loops become relaxed. Cellular DNase introduces nicks into super-coiled loops, leading to the formation of relaxed loops. This process alters the DNA structure, affecting its super-coiling state.